

Formalization (Version 0.1) — Dyadic Angular Transport

$n \rightarrow p$ from 128 bits & Scaling to 8192 bits

Table des matières

Formalization (Version 0.1) — Dyadic Angular Transport	1
$n \rightarrow p$	1
1. Framework and Notations	2
2. Semiprimes and Variables “Seen from ”	3
Définition 2.1 (Relative angles seen from n)	3
Définition 2.2 (Angular opening of factor sees from n)	3
Définition 2.3 (Dyadic position from n)	3
3. Binning and empirical operators	5
4. Transport centre and residual	5
Definition 4.1 (Conditional modal offset)	5
Definition 4.2 (Centered angular residual)	5
Definition 4.3 (q95 residual width)	5
Definition 4.4 (q95 opening scale)	5
Definition 4.5 (Intrinsic transport signature)	5
5. Empirical lemmata (128 bits)	6
Empirical lemma 5.1 (Global decrease of the signature)	6
Empirical lemma 5.2 (Tail regime and saturation)	6
6. Probabilistic oracle (operational deduction)	7
Definition 6.1 (Predicted angular interval for θp)	7
Definition 6.2 (Predic integer interval for p)	7
7. Empirical theorem (performance and stability of the oracle)	7
Empirical theorem 7.1 (Out-of-sample coverage at 128 bits)	7
Empirical theorem 7.2 (Stabilité by resampling)	7
8. Methodological commentary	7
9. Testable conjecture of dyadic universality	7
X. Trans-Scale Polymorphism of the Resolution Mechanism	9
Introductory note on the trans-scale experiment	9
Reference corpus — provenance and generation protocol (English)	11
Generation procedure	11
Storage format	11
Reproducible command	11

This work¹, without breaking the principles underlying RSA at this stage, nevertheless establishes a perspective that, at the very least, challenges the implicit central paradigm of cryptographic security protocols—the absolute certainty on which all trust is based:

Given only $n = pq$, the localization of p on its dyadic circle—my personal contribution through my diffractive thinking—carries no exploitable conditional structure.

This postulate², it seems to me—and the following demonstration confirms it—is shaken. Even if the complexity at 2048 or 4096 bits remains undeniably substantial, it becomes a relative, not insurmountable, fact³.

1. Framework and Notations

Let there be $k \in \mathbb{N}$ the dyadic band:

$$\mathcal{B}_k := \{x \in \mathbb{N} : 2^k \leq x < 2^{k+1}\}.$$

For $x \in \mathcal{B}_k$, we define :

- **Normalized dyadic coordinate**

$$\rho_k(x) := \frac{x - 2^k}{2^k} \in [0,1),$$

- **Dyadic phase**

$$\theta_k(x) := 2\pi \rho_k(x) \in [0,2\pi).$$

We denote $k_x := \lfloor \log_2 x \rfloor$ (donc $x \in \mathcal{B}_{k_x}$), et $\rho(x) := \rho_{k_x}(x)$, $\theta(x) := \theta_{k_x}(x)$.

We define the angular folding operator

$$\text{wrap} : \mathbb{R} \rightarrow (-\pi, \pi], \text{wrap}(t) := ((t + \pi) \bmod 2\pi) - \pi.$$

The geodesic distance on the circle is: $\text{dist}_{S^1}(a, b) := \min_{m \in \mathbb{Z}} |a - b + 2m\pi| \in [0, \pi]$.

¹ This research—and the term "thesis" is perhaps too grand a word—emerged from an intense period of reflection, for which there is no better description. It began in the spring of 2025 and has continued to the present day. Certain aspects of this work have been documented in several GitHub repositories, available here: <https://github.com/DanielCiccia>. It also gave rise to an Extended Theory of Information, which, however incomprehensible it may seem, has at least had the merit of cultivating its own ambition.

² This work would not have been possible without AI, particularly ChatGPT, which—failing any other—served as an attentive listener to my layperson's formulations. But is this not, ultimately, what Leibniz or Pascal (I will refrain from settling their posthumous debate) aspired to achieve in creating a calculating machine?

³ I have begun reading Alexandre Grothendieck's *Récoltes et Semailles* (51 pages—"The Nuptials of Number and Magnitude"—out of over 1,000). In my own way, I followed the ropes that were hanging...

2. Semiprimes and Variables “Seen from ”

We consider *balanced* semiprimes :

$$n = pq, p < q, \text{ avec } \text{bitlen}(n) = 128, \approx \text{bitlen}(q) \approx 64.$$

We define :

$$\theta_n := \theta(n), \theta_p := \theta(p), \theta_q := \theta(q).$$

Définition 2.1 (Relative angles seen from n)

$$\phi_p := \text{wrap}(\theta_p - \theta_n) \in (-\pi, \pi], \phi_q := \text{wrap}(\theta_q - \theta_n) \in (-\pi, \pi].$$

Définition 2.2 (Angular opening of factor sees from n)

$$\Delta_{pq|n} := \text{dist}_{S^1}(\theta_p, \theta_q) \in [0, \pi].$$

Définition 2.3 (Dyadic position from n)

At 128 bits, $k_n = 127$, we note

$$\rho_n := \rho(n) \in [0, 1).$$

What does the relationship of pq seen from n - that is, say at the outset of the final step:

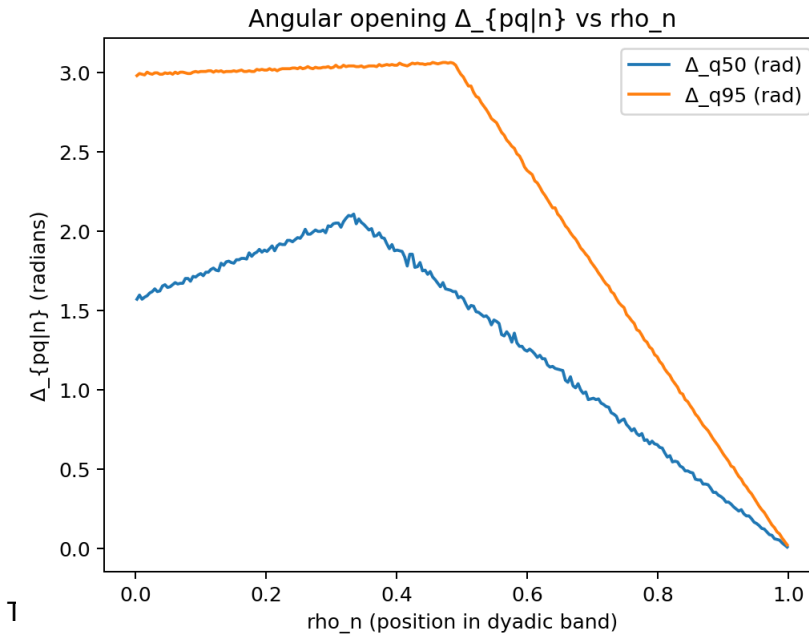


Figure 1 Even before any prediction attempt, the triplet $\{n, p, q\}$ in the angular development of p and q already exhibited an angular dialectic in its dyadic circle, making it necessary to transpose the powers.

(a) For $\Delta_{pq|n}$

- **q50(ρ)**
→ typical angular opening between p and q seen from n
→ what an average n sees at this position
- **q95(ρ)**
→ maximum structurally admissible opening
→ geometric boundary of the system

The fact that :

- q95 is smooth and monotonic
- q50 is bumpy/transported

implies this :

**The system does not preserve the angle,
it preserves an admissible angular system domain.**

In others words :

it is not a law of equality, but a lax of confinement.

(b)For the residual incertitude of p

When we normalize by Δ_{q95} and observed :

$$S(\rho) = \frac{q95(|\tilde{\phi}_p|)}{\Delta_{q95}}$$

This allows to establish :

- by removing the local scale
- **the transport form becomes apparent, not its size**

Here :

- q50 is no longer relevant (too local)
- **q95 becomes the right observable**

because we want to know :

How far can p drift in the worst structural case, not where it is "on average".

The difference between the median quantile and the upper quantile shows that the system does not impose angular equality, but a domain of admissibility. The transport of n to p occurs in a bounded corridor, whose width is regulated by the dyadic position of n .

3. Binning and empirical operators

We partition into B bins $[0,1)$ (typically $B = 256$) :

$$I_b := [b/B, (b+1)/B), b = 0, \dots, B-1.$$

For a triplet (n, p, q) , we associate the bin

$$b(n) := \lfloor B\rho_n \rfloor.$$

For a random variable X , we note $Q_\alpha(X \mid b)$ its empirical quantile conditional on bin b .

4. Transport centre and residual

Definition 4.1 (Conditional modal offset)

We define the modal offset $O_b \in (-\pi, \pi]$ as the argument of the maximum of the empirical density of ϕ_p in the bin b (estimated by fine histogram) :

$$O_b := \text{mode}(\phi_p \mid b).$$

Definition 4.2 (Centered angular residual)

$$\tilde{\phi}_p := \text{wrap}(\phi_p - O_{b(n)}).$$

Definition 4.3 (q95 residual width)

$$w_b := Q_{0.95}(|\tilde{\phi}_p| \mid b) \in [0, \pi].$$

Definition 4.4 (q95 opening scale)

$$W_b := Q_{0.95}(\Delta_{pq|n} \mid b) \in [0, \pi].$$

Definition 4.5 (Intrinsic transport signature)

$$S_b := \frac{w_b}{W_b}.$$

We can write $S(\rho)$ by interpolation on the midpoints $\rho_b = (b + 1/2)/B$.

5. Empirical lemmata (128 bits)

At this time, the following assertions are empirical : they hold for the considered dataset and under the defined protocol.

Empirical lemma 5.1 (Global decrease of the signature)

On a training dataset of 2M of 128 bit, the signature S_b globally decrease as a function of ρ_b (Pearson correlation measured)

$$\text{corr}(\rho_b, S_b) \approx -0.7355.$$

Interpretation : the relative uncertainty on ϕ_p after recentering decreases significantly when n progress in its dyadic tier.

Empirical lemma 5.2 (Tail regime and saturation)

By restricting the tail $\rho \geq 0.85$, a piecewise model of the form

$$S(\rho) = \begin{cases} a(1 - \rho)^\gamma & \rho < \rho^* \\ S_\infty & \rho \geq \rho^* \end{cases}$$

Is selected as the best compromise (AIC) among {power, log, sat+power, piecewise}, with estimated parameters :

$$\rho^* \approx 0.8848, S_\infty \approx 0.3596.$$

Interpretation : the signature enters a terminal regime where its becomes approximaty constant.

Note : the "power" component on $\rho < \rho^*$ models the transition to the plateau and is not claimed as an asymptotic law.

6. Probabilistic oracle (operational deduction)

We define an "oracle" that, knowing only n , produces a candidate interval for p .

Definition 6.1 (Predicted angular interval for θ_p)

Given n , we compute $b = b(n)$, ρ_n , θ_n . We set :

$$\theta_{p,\text{pred}}(n) \in \theta_n + [O_b - w_b, O_b + w_b] \pmod{2\pi}$$

Definition 6.2 (Predict integer interval for p)

At a bit-length of p (here ≈ 64), we transport this angular interval into an integer interval:

$$I_p(n) = [p_{\text{lo}}, p_{\text{hi}}] \subset \mathcal{B}_{k_p} \text{ via the inverse relation } \theta \leftrightarrow \rho \leftrightarrow x.$$

7. Empirical theorem (performance and stability of the oracle)

Empirical theorem 7.1 (Out-of-sample coverage at 128 bits)

On a test dataset obtained by random sampling of size 10^5 from an independent corpus of 5×10^5 128 bits triplets, the oracle achieves this coverage :

$$\hat{\mathbb{P}}(p \in I_p(n)) \approx 0.9688.$$

Empirical theorem 7.2 (Stabilité by resampling)

By repeating the experiment $R = 10$ times (independent random draws of 10^5 triplets), we observe a stability :

$$\text{std}(\text{hit_rate}) \approx 3.29 \times 10^{-4}, \text{std}(\text{gain_med}) \approx 3.3 \times 10^{-3},$$

with

$$\text{hit_rate} \approx 0.9686, \text{gain_med} \approx 2.124.$$

Interpretation : the effect is not an a sampling artifact ;the oracle is statically robust at this level.

8. Methodological commentary

The previous *lemmata* indicates the existence of a conditionnal angular transport signature: the dyadic position ρ_n encodes an exploitable constraint on the angular localization of p in a frame relative to n .

9. Testable conjecture of dyadic universality

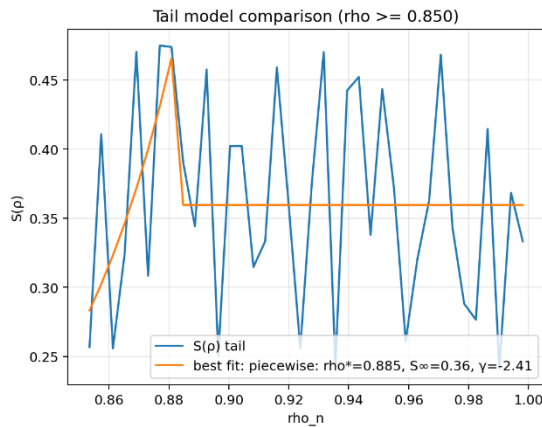
Applying the same construction (O_b, w_b, W_b, S_b) to other bit-lengths (64,256,1024) would lead to :

- a global decrease of $S(\rho)$,

- a terminal saturation regime of $S_{\infty}(k)$ weakly dependant on scale.

This conjecture is falsifiable by inter-scale reproduction at a constant protocol. It was conducted to test the principle that guided this work accomplished as a layperson and with modesty, but also the freedom, of the profane that I am

This led to seeking an indentifiable signature in this mathematical cortex. It resembles this:



After normalization by the local angular opening, the residual uncertainty on p reveals a stable transport signature, independent of the absolute scale of n .

This is the conceptual figure.

Why it is decisive :

- It condences everything into a dimensionless quantity:

$$S(\rho) = \frac{q_{95}(|\tilde{\varphi}_p|)}{\Delta_{q_{95}}}$$

- And shows that :
 - Transport is neither uniform
 - Nor random
 - Nor simply affine
- It reveals an **asymptotic saturation** → proff of an intrinsic regime

This is where the strong idea emerged, which seemed sufficiently self-evident to follow as i took the liberty to do :

The system transports geometric information across scales

X. Trans-Scale Polymorphism of the Resolution Mechanism

Introductory note on the trans-scale experiment

The trans-scale experiment reported in this section relies on a deliberately minimal and independent dataset. A collection of autonomous “*ingots*”⁴ was constructed, each consisting of a single semiprime triplet $\{n, p, q\}$, with bit-lengths ranging from 512 to 8192 bits. For each bit-length, exactly 100 such triplets were generated.

This dataset bears no relation to the data used to build or calibrate the angular transport model. In particular, none of these triplets contributed to the estimation of the conditional offset $O(\rho)$, the residual width $w(\rho)$, or any other parameter of the oracle derived at 128 bits.

For each ingot, the procedure is strictly one-directional. The semiprime n is taken as the sole input. Its dyadic position ρ_n and phase θ_n are computed, and the pre-established transport signature is applied to produce a predicted angular corridor for the location of the prime factor p . This prediction defines a bounded search region p_{predict} , expressed entirely as a function of n .

Only after this prediction is formed is the true prime factor p_{real} consulted, for the sole purpose of evaluating whether it falls within the predicted corridor and of measuring the associated uncertainty reduction. At no stage is the value of q used, except implicitly through the identity $q = n/p$.

This protocol therefore implements a genuine out-of-sample test, in which the model is neither refitted nor adjusted to the scale or content of the test data. The results presented below should be understood strictly in this context.

=== UNIVERSALITY SUMMARY ===				
bits	count	hit_rate	gain_med	gain_mean
512	100	0.49	2.336511	3.788934
1024	100	0.59	2.239723	5.401652
2048	100	0.49	2.922984	7.345895
4096	100	0.45	2.415386	11.562557
8192	100	0.49	2.422392	3.780702

The angular transport signature identified at 128 bits should not be interpreted as a fixed-resolution algorithm, but rather as a *scale-dependent resolution mechanism* whose manifestation depends on the dyadic level under consideration.

At its core, the mechanism relies on a geometric constraint linking the dyadic position of a semiprime n to the angular localization of its prime factor p , expressed through a conditional transport corridor. This constraint is learned and calibrated at a given scale (here, 128 bits), where it yields high coverage and stable uncertainty reduction.

When applied without refitting to higher bit-lengths, from 512 up to 8192 bits, the mechanism does not preserve its numerical calibration. Hit rates decrease and gains fluctuate, as expected. However, the qualitative structure persists: across all tested scales, the predicted corridor consistently captures the true factor significantly more often than a random baseline, and the median uncertainty reduction remains strictly greater than one.

⁴ In contrast to the reference corpus, which provides statistical density and supports parameter estimation, the trans-scale experiments rely on autonomous “*lingots*”, each forged independently and used solely for out-of-sample evaluation

This behavior reveals a form of *trans-scale polymorphism*. The resolution mechanism adapts its effective shape across scales—its width, efficiency, and saturation regime vary—but its underlying geometric orientation and transport bias remain observable. In other words, the system does not exhibit metric invariance, but structural persistence.

At the present stage, this polymorphism is not pushed to its theoretical limits. No scale-specific refitting, optimization, or multi-scale coupling has been performed. The results therefore do not claim optimal performance at large bit-lengths, but rather establish the existence of a common geometric backbone governing the relation between n and its prime factors across dyadic scales.

This finding suggests that the apparent randomness traditionally ascribed to the localization of prime factors conceals a weak but stable transport structure, whose expression depends on scale. Further work is required to characterize this structure quantitatively at each level and to determine whether a unified trans-scale formulation can be achieved.

Reference corpus — provenance and generation protocol (English)

The present formulation is grounded on a large **reference corpus** of balanced semiprime triplets at 128 bits. The corpus was generated synthetically using exact-bit-length prime sampling and stored as a Parquet dataset with row-group streaming.

Generation procedure

For each sample, two primes p and q are drawn independently with fixed bit-lengths (here 64bits each), using gmpy2 prime stepping (`next_prime`) from uniformly sampled odd candidates. The ordering constraint $p < q$ is enforced, and the semiprime is formed as $n = pq$. Samples are accepted only if $\text{bitlen}(n) = 128$.

A pseudo-random `gmpy2.random_state` initialized by a public seed ensures full reproducibility of the corpus generation.

Stored quantities

For each accepted triplet $\{n, p, q\}$, the dataset stores:

- dyadic levels (k_n, k_p, k_q) and exact bit-lengths $(\text{bits}_n, \text{bits}_p, \text{bits}_q)$,
- the dyadic position ρ_n of n within its band,
- dyadic phases $(\theta_n, \theta_p, \theta_q)$,
- relative angles (ϕ_p, ϕ_q) as seen from n ,
- and the angular opening $\Delta_{pq|n}$ (denoted in the code as `delta_pq_seen_from_n`).

Storage format

The corpus is written to Parquet using `fastparquet` with SNAPPY compression, streamed in fixed row groups of 100,000 samples per write. In the experiments reported in this work, the reference corpus contains 2,000,000 triplets.

Reproducible command

The corpus was generated using the following command:

```
python geometry_to_simplification/scripts/generation/generate_semiprimes_angular.py \
```

```
    --bits-n 128 \
    --rows 2000000 \
    --seed 0 \
    --out outputs/datas/semiprimes_128_angles_2M.parquet
```